

Geometry
Unit 1
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Line segment KN is shown where $KR = 16x - 3$, $RW = WN$, $WN = 9x - 3$, $BR = \frac{1}{2}RW$, and $KN = 161$.



1. Find the value of x .

$$\begin{aligned}16x - 3 + 9x - 3 + 9x - 3 &= 161 \\34x - 9 &= 161 \\34x &= 170 \\x &= 5\end{aligned}$$

2. Find the length of $\overline{RW}$.

$$9 \cdot 5 - 3 = 45 - 3 = 42$$

3. Find the length of $\overline{KW}$.

$$\begin{aligned}16 \cdot 5 - 3 &= 80 - 3 = 77 \\77 + 42 &= 119 \\119\end{aligned}$$

4. Find the length of $\overline{BR}$.

$$\frac{42}{2} = 21$$

5. Use a compass and straightedge to make a copy of $\overline{HT}$.



T , M , W , and H are four collinear points that exist such that $\overline{TW}$ and $\overline{MH}$ are congruent. $TW = 5y + 10$ and $MH = 9y - 46$. Four different students draw a diagram of the description. Each student's drawing is shown.

Name	Drawing
Miquel	
Harper	
Camila	
Liam	

6. Determine the value of y .

$$\begin{aligned}
 5y + 10 &= 9y - 46 \\
 56 &= 4y \\
 14 &= y
 \end{aligned}$$

7. Use Miquel's drawing to determine the length of $\overline{WM}$ if $\frac{7}{3}WM = MH$.

$$\begin{aligned}
 MH &= 9y - 46 = 9 \cdot 14 - 46 = 80 \\
 \frac{7}{3}WM &= MH; \quad \frac{7}{3}WM = MH; \quad \frac{7}{3}WM = 80 \\
 WM &= 80 \cdot \frac{3}{7} = \frac{240}{7} = 34\frac{2}{7}
 \end{aligned}$$

8. Use Harper's drawing to determine the length of $\overline{MW}$ if $\frac{3}{5}TM = MW$.

$$\begin{aligned}
 TM + MW &= TW; \quad TM + \frac{3}{5}TM = 5 \cdot 14 + 10; \quad \frac{8}{5}TM = 80 \\
 TM &= 50; \quad MW = 80 - 50 = 30
 \end{aligned}$$

9. Use Camila's drawing to determine the length of $\overline{MW}$ if $TM + MW + WH = 130$.

$$80 + WH = 130; \quad WH = 50; \quad 130 - 100 = 30 = MW$$

10. Use Liam's drawing to determine the length of $\overline{WM}$ if $TH = 190\frac{1}{3}$.

$$\begin{aligned}
 80 + x + 80 &= 190\frac{1}{3} \\
 x &= 30\frac{1}{3}
 \end{aligned}$$